

Regulation of Plasma Glucose

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Block: Endocrine/Urology/Reproduction
Thread: Biochemistry
Conflict of Interest: Dr. Scott has no conflicts of interest to disclose.

Learning objectives

1. Summarize the mechanism of glucose sensing by pancreatic beta-cells.
2. Compare the insulin sensitivities of lipolysis, glucose production and glucose disposal.
3. Outline maturation of peptide hormones using insulin as an example. Explain the source of C-peptide and the clinical utility of C-peptide assays.
4. Describe the normal physiological counter-regulation to insulin-induced hypoglycemia.
5. Relate the structures, functions and regulation of glucagon, GIP and GLPs and explain how GIP and GLPs enhance insulin secretion.

Practice questions

1. **Chronic hypokalemia** can lead to **diabetes**. Provide a likely mechanism that explains this finding.
2. Outline the events that take place during lipolysis (name the tissue(s), the starting materials, the regulated enzymes, the regulators, and the products).
3. What are the similarities and differences in the major stimuli for glycogen synthesis in the liver and muscle?
4. Sketch out the sequence of molecular events that connect glucagon in the extracellular space to glycogenolysis inside a hepatocyte.
5. A 6-year-old boy suffered from severe hypoglycemia upon prolonged (>24 hr.) fasting. In the liver, the conversion of glucose to lactate was normal. The increase in blood glucose following administration of glucagon was nearly normal (the individual was in a well-fed state prior to this test). Analysis of a liver biopsy showed normal amounts of pyruvate carboxylase and phosphoenolpyruvate carboxykinase activities. Assuming that this is a heritable disorder, propose an enzyme defect that accounts for these findings.

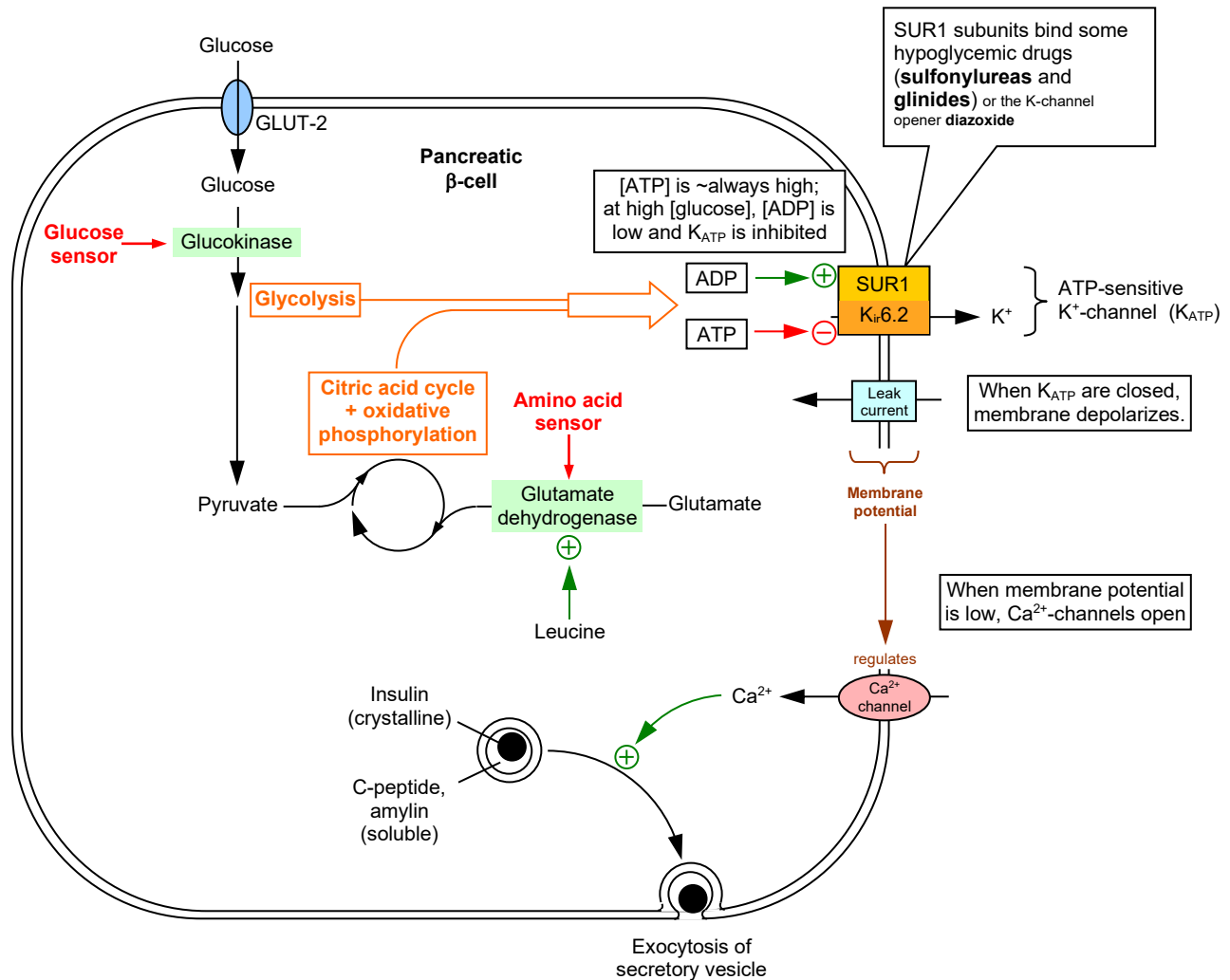
Introduction

Plasma glucose concentration is regulated primarily through the action of pancreatic and adrenal hormones on adipose, liver and muscle, with additional regulatory inputs from intestinal hormones. Insulin regulates the fed state through inhibition of fat utilization (lipolysis) and endogenous glucose production (glycogenolysis and gluconeogenesis), and stimulation of glucose uptake, storage and utilization through activation of glycogen synthesis and glycolysis. In the fasting state, glucagon and epinephrine stimulate lipolysis, glycogenolysis and gluconeogenesis.

A. Glucose Sensing by Pancreatic β -cells

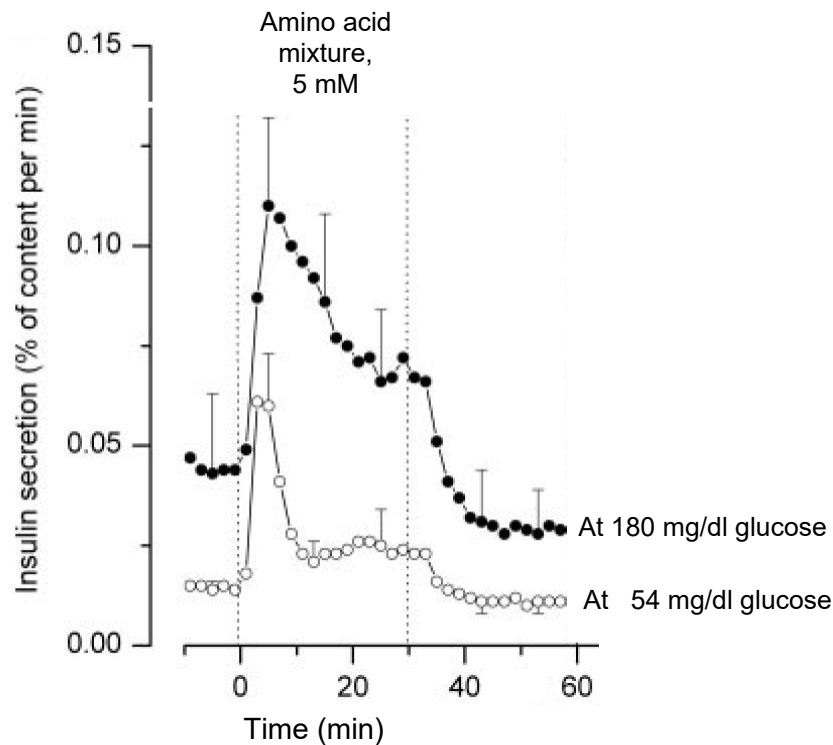
1. Pancreatic β -cells secrete **insulin** in response to an elevated concentration of glucose. Amino acids, fatty acids, and ketone bodies can enhance glucose-induced insulin secretion, while epinephrine inhibits it.
2. Insulin secretion is currently thought to be regulated by glucose as follows:
 - a. **Glucokinase** (which catalyzes the reaction $\text{glucose} + \text{ATP} \rightarrow \text{glucose 6-phosphate}$) is an integral part of the glucose sensor in pancreatic β -cells. The glucose-dependence of glucokinase activity determines the glucose sensitivity of insulin secretion. In β -cells, glucokinase also determines flux through glycolysis.
 - b. β -cells are electrically **excitable**, like neurons.
 - c. β -cells and most other cells almost always have a high concentration of **ATP** that changes very little. By contrast, the concentration of free ADP changes appreciably (recall that oxidative phosphorylation is limited by ADP concentration – see Overview of Energy Metabolism lecture in the Cardio block).
 - d. In the **absence** of glucose, an elevated concentration of **ADP** allows **K_{ATP}-channels** (ATP-sensitive K-channels) to open; thus, K⁺ flows out of β -cells and hyperpolarizes them so that they **cannot** secrete insulin.

- e. In the **presence** of an elevated concentration of **glucose**, ADP is mostly converted to ATP, which keeps the K_{ATP} mostly **closed**.
- i. Leak channels allow β -cells to depolarize. Ca^{2+} -channels then open and allow Ca^{2+} into the β -cell, where Ca^{2+} activates exocytosis of insulin-containing secretory granules.



- f. **Sulfonylureas and glinides have the same effect as glucose: they close K_{ATP} , thus stimulating insulin secretion.** These drugs stimulate insulin secretion regardless of the concentration of glucose, and **hypoglycemia** is therefore one of their side-effects.

3. **Amino acids** (especially **leucine** and **glutamine**) **enhance** glucose-induced insulin secretion.



Amino acids enhance glucose-induced insulin secretion from isolated human pancreatic islets. The islets were perfused in vitro. The amino acid mixture was based on physiological concentrations. Slightly modified from J-C Henquin et al., Diabetes 2006;55:3470-3477.

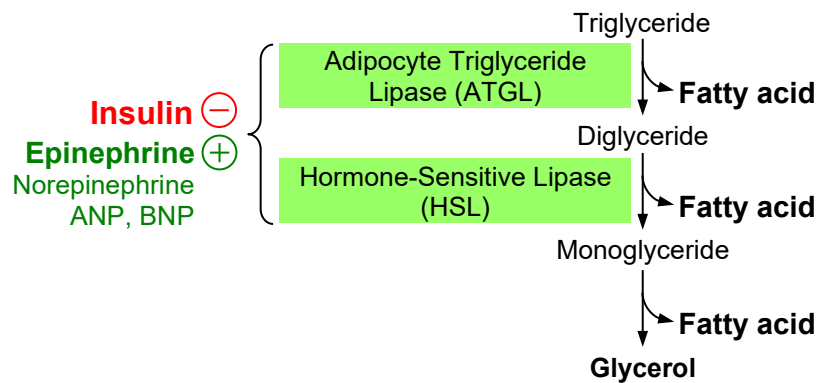
- a. **Glutamate dehydrogenase** (see center of diagram above), which catalyzes the reaction $\text{glutamate} \leftrightarrow \alpha\text{-ketoglutarate} + \text{NH}_3$, is an **amino acid sensor**.
 - b. Physiologically, the concentration of **amino acids** is fairly **constant** (it is maintained jointly by insulin and glucagon; insulin stimulates use of amino acids in protein synthesis, while glucagon favors use of amino acids in gluconeogenesis).
4. **Fatty acids** and **ketone bodies** also enhance glucose-induced insulin secretion.
- a. After a **meal**, this enhancement is essentially irrelevant, because the effects of glucose and amino acids on insulin secretion are far greater.
 - b. During a **fast**, stimulation of insulin secretion by fatty acids and ketone bodies is **crucial**, because insulin is the only hormone that attenuates **lipolysis** and thus keeps us from producing more ketone bodies than we

can use. The insulin-dependent attenuation of lipolysis prevents us from developing **ketoacidosis** in the fasting state (see section B).

- c. **We need insulin at all times.** After a meal, we use insulin to prevent hyperglycemia. In the fasting state, we use insulin to prevent ketoacidosis. Accordingly, patients who have type 1 diabetes inject a long-acting insulin or use a basal rate of an insulin infusion, and for meals, they supplement this with a rapid-acting injected or infused insulin (see PHARM, Week 2, Type 1 Diabetes & Insulin).

B. Effect of Insulin on Lipolysis

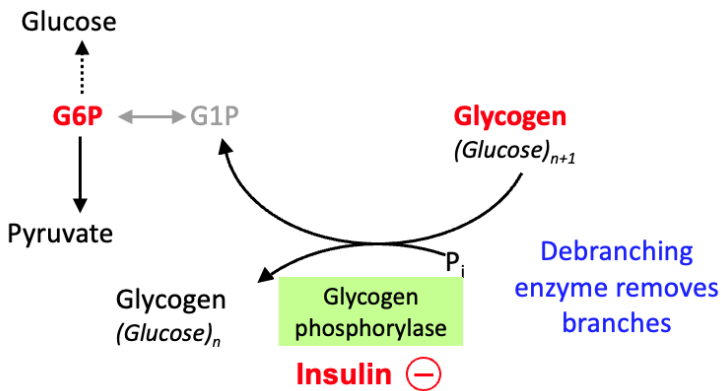
1. Lipolysis refers to the **hydrolysis of triglycerides inside adipocytes** (see Cardio Block, Week 1: Lipid Metabolism).
2. Lipolysis yields **fatty acids** and **glycerol**; both are released into the blood:



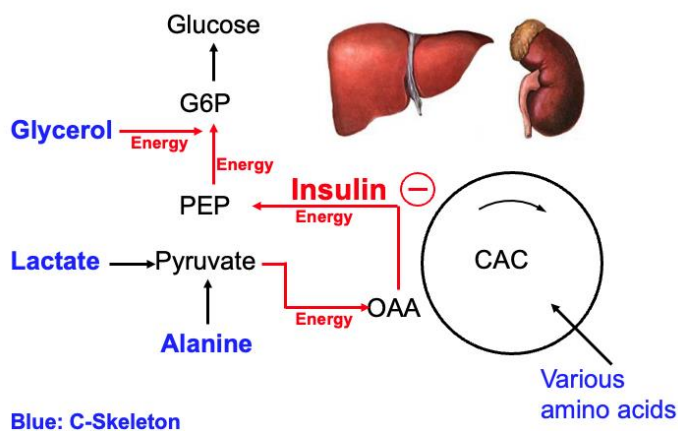
3. Lipolysis is highly sensitive to inhibition by insulin. Insulin regulation ensures that lipolysis only occurs during **fasting** and **exercise**.
4. β -Oxidation of fatty acids yields **acetyl-CoA**. The liver can turn some of this acetyl-CoA into **ketone bodies** (**acetoacetate** and **β -hydroxy-butyrate**). Insulin inhibition of lipolysis is critical to prevent ketoacidosis.

C. Effect of Insulin on Endogenous Glucose Production

1. **Endogenous glucose production** results from the combined activities of **glycogen phosphorylase** in the liver, which liberates glucose from hepatic glycogen stores, and **gluconeogenesis**, which carries out energy-dependent synthesis of glucose from amino acids, glycerol or lactic acid. Endogenous glucose production is moderately sensitive to inhibition by insulin.



2. **Glycogen phosphorylase** controls glycogen degradation.
 - a. The **liver** hydrolyzes glycogen to **glucose** in the **fasting state** and during moderate to intense **exercise**.
 - i. Insulin inhibits glycogen phosphorylase in the liver and muscle.
3. The regulation of gluconeogenesis works as follows:

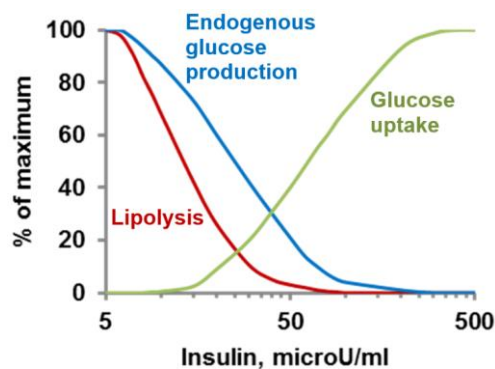


- a. **Phosphoenolpyruvate carboxykinase (PEPCK)** is the primary rate-controlling enzyme of gluconeogenesis.
 - i. **Insulin** inhibits PEPCK.

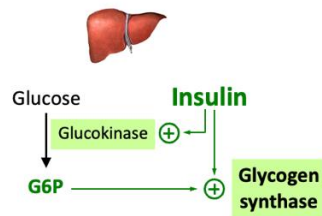
- b. Gluconeogenesis is also inhibited when the concentration of **AMP** is high (indicating lack of ATP synthesis from fatty acid β -oxidation; red arrows above indicate reactions that require energy).

D. Effect of Insulin on Glucose Utilization

1. **Glucose utilization** results from an insulin-dependent increase in **glycogen synthase** activity and stimulation of **glycolysis**. Insulin activation of glucose utilization requires a higher concentration of insulin than inhibition of lipolysis or endogenous glucose production.

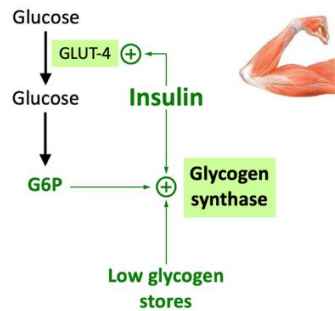


2. **Glycogen synthase** controls glycogen synthesis.



- In the **liver**, glycogen synthesis occurs in the **fed state**.
 - Glucokinase** converts glucose to **glucose 6-phosphate**.
(Unlike hexokinase, glucokinase is not inhibited by glucose-6-phosphate)
Glucokinase activation also stimulates **glycolysis**.
 - Glucose 6-phosphate** stimulates glycogen synthase.

iii. **Insulin** stimulates glucokinase and glycogen synthase.

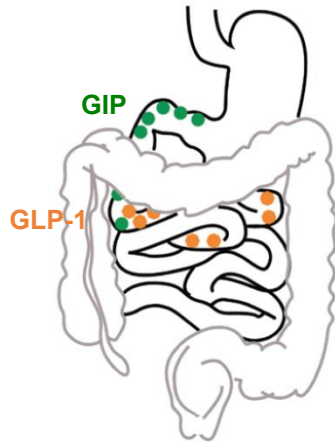


- b. In **skeletal muscle**, glycogen synthesis occurs **after exercise**: at a moderate pace, if the person is fasting, and at a considerably increased rate **after a meal**. In muscle (as in liver), the rate of glycogen synthesis depends on glycogen synthase activity. Muscle glycogen synthase differs from liver glycogen synthase (these enzymes are encoded by different genes).
 - i. **Low glycogen stores** stimulate glycogen synthase (mechanism unknown).
 - ii. **Insulin** stimulates insertion of insulin-sensitive (**GLUT-4**) glucose transporters into the plasma membrane of adipose and muscle. Increased surface area for glucose absorption through GLUT-4 availability contributes to glucose uptake.
 - iii. Insulin stimulates **glycogen synthase** in muscle.

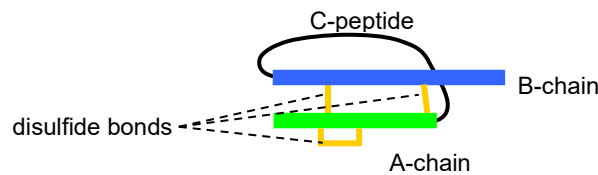
E. Insulin, Glucagon, GLP-1, and GIP as Regulators of Plasma Glucose

1. **Pancreatic islets** make up about 1% of the volume of the pancreas, are dispersed in the pancreas, and contain:
 - a. **α -cells** that synthesize and secrete **glucagon**.
 - b. **β -cells** that synthesize and secrete **insulin**.
 - c. Other cells
2. When the intestine senses food, it secretes **incretins**, i.e., hormones that potentiate insulin secretion.
 - a. **K-cells** in the **intestine** synthesize and secrete **gastric inhibitory peptide (GIP)**, which is also called glucose-dependent insulinotropic polypeptide (same abbreviation).

- b. **L-cells** in the **intestine** synthesize and secrete **glucagon-like peptide 1 (GLP-1)**.



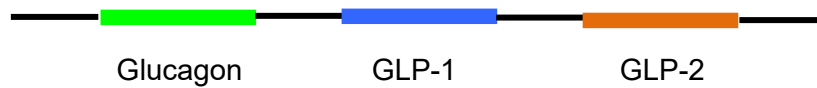
3. Insulin, glucagon, GIP, and GLP-1 are small **peptides**.
- These peptides result from proteolysis of longer precursors, whereby glucagon and GLP-1 derive from the same precursor (albeit in different cells!):
 - During translation of the mRNA, the growing peptide chain is inserted into the endoplasmic reticulum, and the **signal sequence** (also called **prepro** sequence or **leader** sequence) is cleaved, yielding a **propeptide** (**proinsulin**, **proglucagon**) in the lumen of the E.R. (see also lesson on Membranes and Membrane Proteins in Week 1 of Introduction to the Study of Medicine)
 - The propeptides travel through the **Golgi** apparatus and end up inside **secretory granules**, along with processing enzymes.
 - Inside secretory granules, proinsulin is processed to **C-peptide** and **insulin** (which consists of an **A-chain** and a **B-chain**):



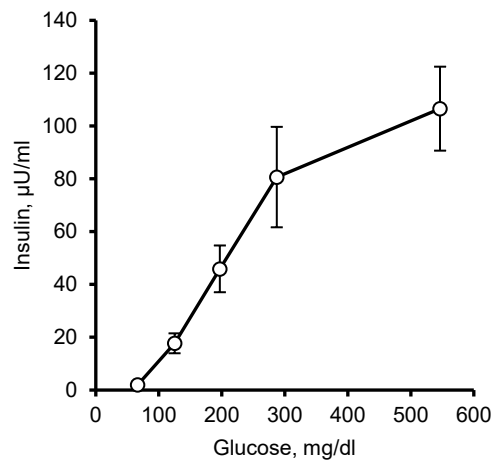
- β -cells store about **one week's** worth of insulin.
- β -cells secrete both insulin and C-peptide.
 - **Synthetic insulin** does not contain C-peptide. Therefore, measurement of the concentration of **C-peptide** in the serum allows one to **distinguish exogenous insulin from endogenous insulin** secretion (e.g., from an insulinoma, or from remaining β -cells in diabetes).

➤ C-peptide is different from C-reactive protein (CRP).

- iv. Inside secretory granules of pancreatic α -cells, proglucagon is processed to **glucagon**.
- v. Inside secretory granules of L-cells in the intestine, proglucagon is processed to **GLP-1** (and GLP-2).

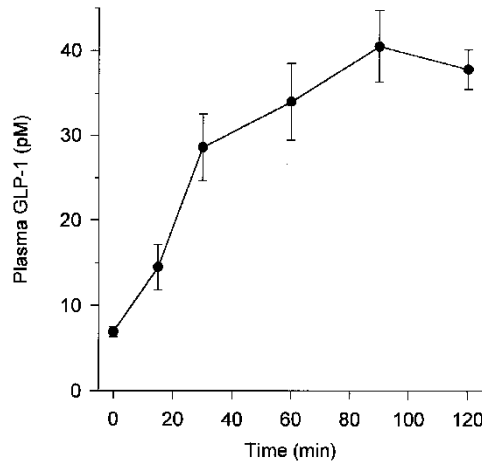


4. **Glucose** is the main and *sine qua non* stimulus for insulin secretion. **Below a plasma glucose of about 35 mg/dl, pancreatic β -cells do not secrete insulin.**



Glucose and Insulin in the Blood (Dose-Response Curve). Glucose was injected into a brachial vein of healthy volunteers. A bolus was followed by a continuous infusion. Ten minutes after the injection of the bolus, blood samples were removed from a vein of the other arm. The concentrations of glucose and insulin in the samples were then determined. (Adapted from: E Cerasi et al., Diabetes 21: 224-234, 1972)

5. **L-cells in the intestine** secrete **glucagon-like peptide-1 (GLP-1)** when they sense carbohydrates and fatty acids:



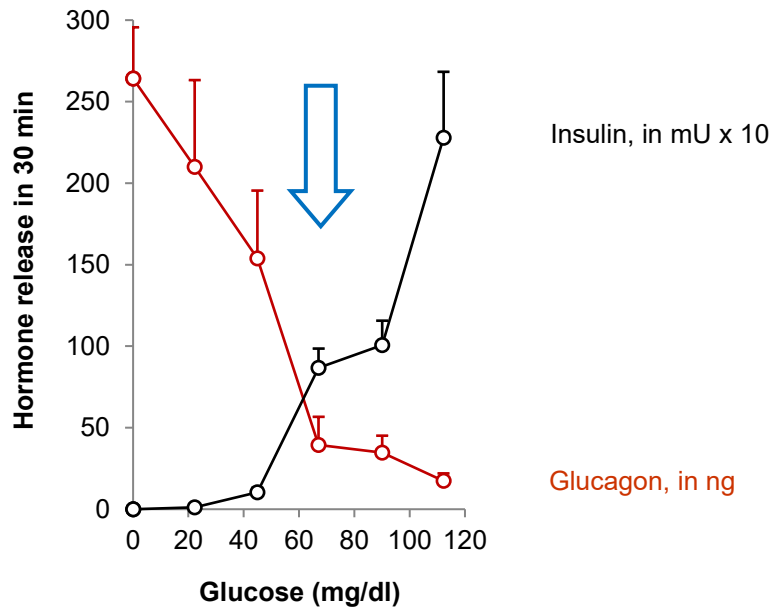
Six volunteers ate a carbohydrate-rich breakfast. Blood was sampled from an antecubital vein.
(Adapted from C Orskov et al., Diabetes 43: 535-539, 1994.)

- Likewise, glucose and fatty acids in the lumen of the intestine increase GIP secretion into the blood.
 - GIP and GLP-1 potentiate insulin secretion at plasma glucose greater than about 100 mg/dl** but not below this concentration. **Note:** the glucose dependence of incretin activity protects against incretin-induced hypoglycemia.
 - GIP receptor agonists, GLP-1 receptor agonists, and inhibitors of the enzyme DPP-4** (which degrades GIP and GLP-1) are used to treat **type 2 diabetes** (PHARM, Management of Type 2 Diabetes in Week 2).
6. GLP-1 and GIP also act on receptors in the stomach and brain, thereby decreasing gastric emptying, appetite and food intake. Consequently, incretins are now also approved for weight management.

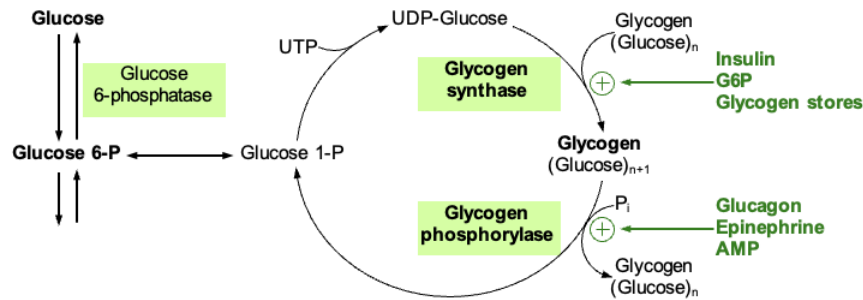
F. Insulin Counter-Regulation

- The concentration of **glucose** in blood is regulated primarily by the concentrations of glucagon and insulin. **We secrete glucagon and insulin at the same time**, largely dependent on the concentration of **glucose**:
 - When the concentration of glucose in plasma is **decreasing**, we secrete more **glucagon** and less insulin.
 - When the concentration of glucose in plasma is **increasing**, we secrete less glucagon and more **insulin**. (The figure below is based on in vitro perfused rat pancreata; n=3).

We live "in the well" between
the extremes of glucagon
and insulin secretion

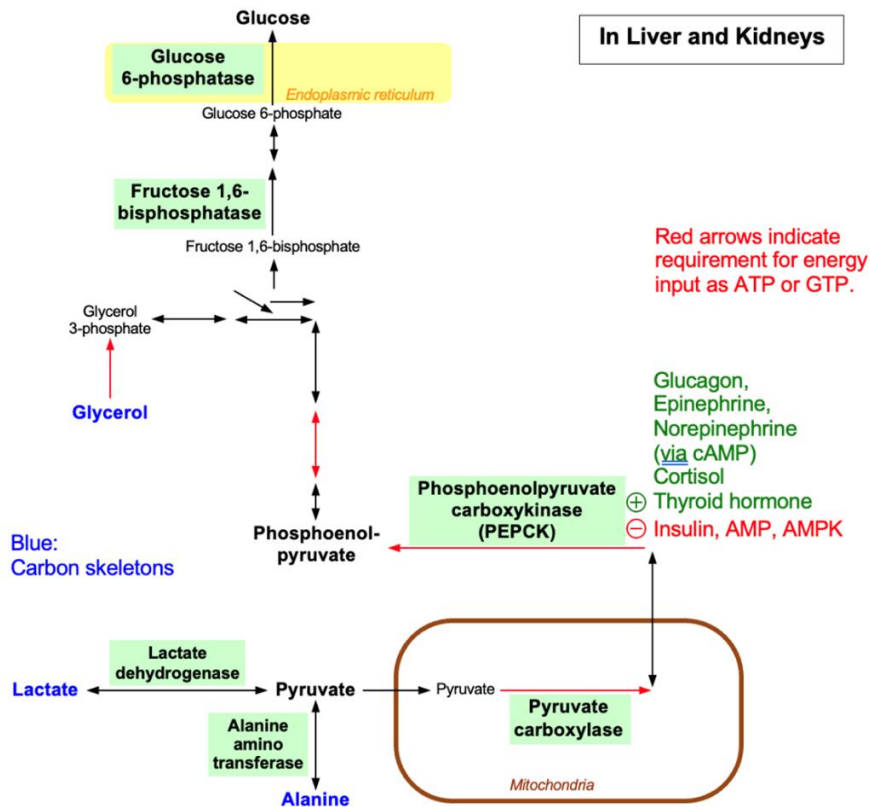


- c. After a meal, when a significant amount of **amino acids** enter the blood, we secrete more glucagon and more insulin. The insulin favors incorporation of amino acids into **body protein**, and the glucagon prevents insulin-induced hypoglycemia.
- d. In the fasting state or during exercise, an increased concentration of glucagon activates **gluconeogenesis**, so that glucogenic amino acids are turned into glucose.
2. Pancreatic α -cells secrete **glucagon** in response to amino acids. Epinephrine enhances amino acid-induced glucagon secretion, while glucose inhibits it.
3. **Epinephrine**, acting via α -adrenergic receptors, is the second major physiological **inhibitor** of insulin secretion.
 - a. Epinephrine limits insulin secretion during **exercise**. (Note: insulin is still secreted during exercise, and insulin secretion rises sharply after intense exercise.)
4. Lipolysis is stimulated by **epinephrine, norepinephrine, ANP** and **BNP**. Epinephrine overrides insulin inhibition of lipolysis.
5. **Glucagon** and **epinephrine** both stimulate glycogenolysis by stimulating glycogen phosphorylase activity in the liver.



6. **Skeletal muscle** hydrolyzes glycogen to **glucose 6-phosphate** during **exercise**.
 - i. Intracellular Ca^{2+} and **AMP** (from the hydrolysis of ATP) are the chief stimuli of glycogen phosphorylase in skeletal muscle.
 - ii. **Epinephrine**, via adrenergic receptors, also stimulates glycogen phosphorylase in skeletal muscle.
 - iii. Skeletal muscle does **NOT** have glucagon receptors.

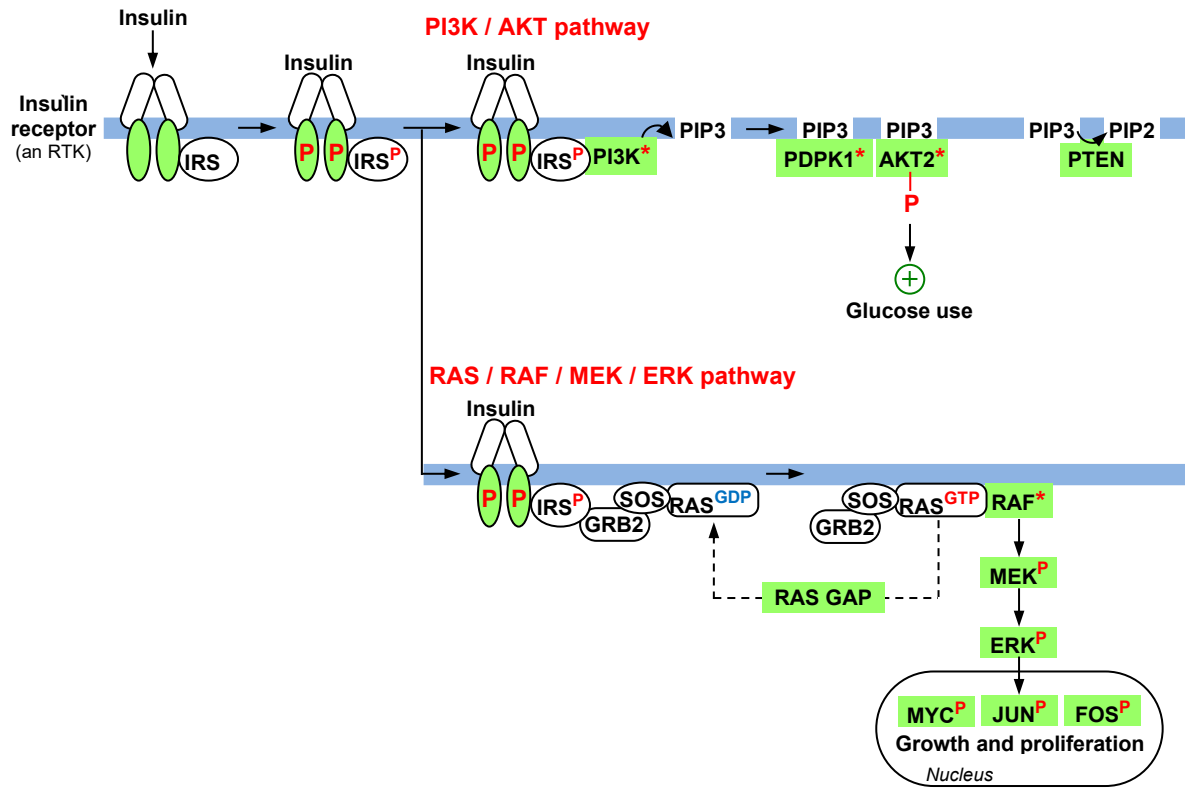
7. **Glucagon** and **epinephrine** activate PEPCK, which stimulates gluconeogenesis.



8. **Glucocorticoids** promote autophagy, which provides amino acids that stimulate glucagon production and provide carbon skeletons for gluconeogenesis.

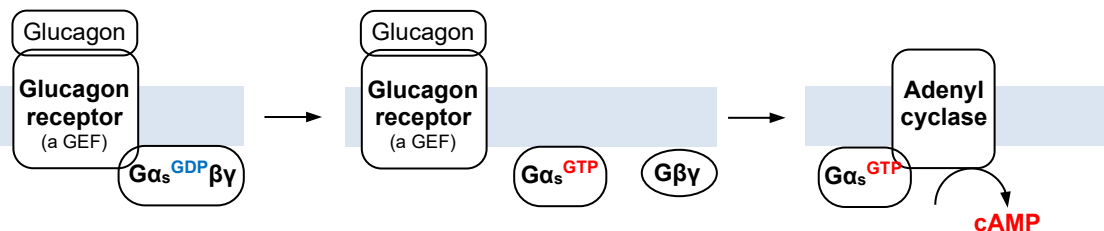
G. Insulin Receptors, Glucagon Receptors, GIP receptors, and GLP-1 Receptors

1. Insulin binds to **insulin receptors**, which are **growth factor receptors** (also known as **receptor tyrosine kinases**) that signal via the **AKT pathway** and via the **MAP kinase cascade** (i.e., the RAS/RAF/MEK/ERK pathway) [see Dr. Wedegaertner's lesson in the Pulmonary Block, Week 5, "Signaling Pathways in Solid Tumor Carcinogenesis", Section A]



- a. In **muscle** and **adipose tissue**, insulin stimulates the insertion of **GLUT-4 insulin-sensitive glucose transporters** into the plasma membrane. This allows muscle to use glucose to synthesize glycogen, and it allows adipose tissue to produce glycerol 3-phosphate, which is esterified with incoming fatty acids, producing triglycerides for deposit. GLUT-4 insertion also dramatically increases surface area for glucose absorption, which enhances insulin-dependent glucose uptake.
- b. In all cells that have a significant number of insulin receptors, insulin tends to **favor glycolysis, glycogen synthesis, protein biosynthesis, and triglyceride synthesis**.
- c. Note that the following two tissues do NOT respond to insulin:

- i. Mature **red blood cells** do not have a significant number of insulin receptors.
- ii. Most **brain** tissue is behind the blood-brain barrier and is therefore **NOT** exposed to insulin
- d. In the liver, insulin **inhibits glycogenolysis**. In the liver and in the kidneys, insulin inhibits **gluconeogenesis**.
- e. Insulin is a **growth factor**, and many cells need insulin in order to replicate.
 - i. **Insulin analogs** used in the treatment of **diabetes** have variable ratios of potency to stimulate/inhibit metabolism vs. cell growth; this may affect **tumorigenesis**.
- 2. **Glucagon receptors** are **G protein-coupled receptors (GPCRs)**; see Signal Transduction in Introduction to the Study of Medicine, Week 2), whereby the heterotrimeric G protein that binds to the receptor contains **G α_s** , which leads to increased **adenyl cyclase** activity and an increased concentration of **cAMP**. cAMP then stimulates **protein kinase A (PKA)**, which **phosphorylates** target proteins.



- a. Glucagon receptors exist mainly in the **liver**.
 - i. In the liver, glucagon leads to increased **glycogenolysis** and increased **gluconeogenesis**.
 - ii. Note that pancreatic β -cells seem to be devoid of glucagon receptors.
- 3. **GIP receptors** and **GLP-1 receptors**, like glucagon receptors, are **GPCRs** that couple to **G α_s** , which leads to an increased intracellular concentration of **cAMP**.
 - a. GIP receptors and GLP-1 receptors are found in several tissues.
 - i. GIP and GLP-1 **potentiate glucose-induced insulin secretion** from **pancreatic β -cells**.
 - ii. GLP-1 **reduces the rate of emptying** of the **stomach**.
 - iii. GIP and GLP-1 induce **satiety**.
 - b. Under physiological circumstances, glucagon receptors only respond to glucagon and GLP-1 receptors only to GLP-1, i.e., these receptors do **NOT** show cross-reactivity.

Answers to practice questions

1. **Chronic hypokalemia** can lead to **diabetes**. Provide a likely mechanism that explains this finding.

Answer: The most likely mechanism for diabetes due to chronic hypokalemia is inadequate insulin secretion due to excessive hyperpolarization of β -cells. Just like neurons, β -cells are hyperpolarized when the K^+ -concentration gradient from the inside of the cell to the extracellular space is increased. (By contrast, hyperkalemia would stimulate insulin secretion.)

2. Outline the events that take place during lipolysis (name the tissue(s), the starting materials, the regulated enzymes, the regulators, and the products).

Answer: The term 'lipolysis' specifically refers to the hydrolysis of triglycerides that are stored in lipid droplets inside adipocytes. ATGL and HSL hydrolyze triglycerides to diglycerides and then monoglycerides. These enzymes are activated by ANP, BNP, and epinephrine (and a little bit also by glucagon). The product of lipolysis is free fatty acids (FFA), which are released into the blood, where they bind to albumin. The adipose tissue also releases glycerol, which the liver converts to glycerol 3-phosphate. Glycerol 3-phosphate, after conversion to dihydroxyacetone phosphate, can enter gluconeogenesis or glycolysis.

3. What are the similarities and differences in the major stimuli for glycogen synthesis in the liver and muscle?

Answer: The liver deposits glycogen mostly after a **meal** in response to an increased concentration of insulin. Skeletal muscle deposits glycogen at a relatively low rate after **exercise** and at a markedly higher rate after a **meal** (stimulated by insulin).

Glucose 6-phosphate stimulates glycogen synthase in both tissues.

The **liver** contains glucose transporters at all times. In the liver, glucose, fructose (from food), and insulin all lead to increased **glucokinase** activity and hence an increased concentration of glucose 6-phosphate. In **muscle**, insulin stimulates the insertion of **GLUT-4** insulin-sensitive glucose transporters into the plasma membrane; this leads to an increase in the concentration of glucose and glucose 6-phosphate in muscle.

4. Sketch out the sequence of molecular events that connect glucagon in the extracellular space to glycogenolysis inside a hepatocyte.

Answer: Glucagon binds to the glucagon receptor, which is a G protein-coupled receptor (GPCR). Glucagon binding to the receptor stimulates guanine nucleotide exchange in the alpha subunit ($G\alpha_s$) of the associated heterotrimeric G protein. GTP bound $G\alpha_s$ dissociates from the receptor (and from $\beta\gamma$), and activates adenyl cyclase, which produces cAMP. cAMP activates protein kinase A (PKA), which activates glycogen phosphorylase through a kinase cascade. [PKA phosphorylates and activates a kinase that phosphorylates and activates glycogen phosphorylase].

Insulin stimulates **glycogen synthase** activity in both muscle and liver.

5. A 6-year-old boy suffered from severe hypoglycemia upon prolonged (>24 hr.) fasting. In the liver, the conversion of glucose to lactate was normal. The increase in blood glucose following administration of glucagon was nearly normal (the individual was in a well-fed state prior to this test). Analysis of a liver biopsy showed normal amounts of pyruvate carboxylase and phosphoenolpyruvate carboxykinase activities. Assuming that this is a heritable defect, propose an enzyme defect that accounts for these findings.

Answer: After 24 hours of fasting, glycogen supplies would be exhausted, and gluconeogenesis would become important. Since problems appeared only after glycogen was depleted, it is likely that there is a defect in gluconeogenesis, and not in glycogen utilization. The conversion of glucose to lactate was normal, indicating that the enzymes of glycolysis are normal. Therefore, the defect must be in one of the irreversible reactions that are unique to gluconeogenesis. Tests for phosphoenolpyruvate carboxykinase and pyruvate carboxylase were normal. Glucose-6-phosphatase must be normal for glycogen utilization to occur during the first 24 hours. This leaves fructose-1,6-bisphosphatase as the most likely defective enzyme.